Student : Zixuan Qian

My task was to implement and test the appointments API in the existing Lab 4 project.

I completed the JavaScript practice exercises, covering variables, arrays, objects, functions, modules, error handling and asynchronous code.

I implemented two endpoints: GET /api/appointments, which returns the appointment list with HTTP 200, and POST /api/appointments, which creates an appointment with HTTP 201 and a unique ID. The required fields are doctor, patient and date. Missing, empty or non-string fields return HTTP 400 without adding an appointment. I also enabled JSON body parsing using express.json().

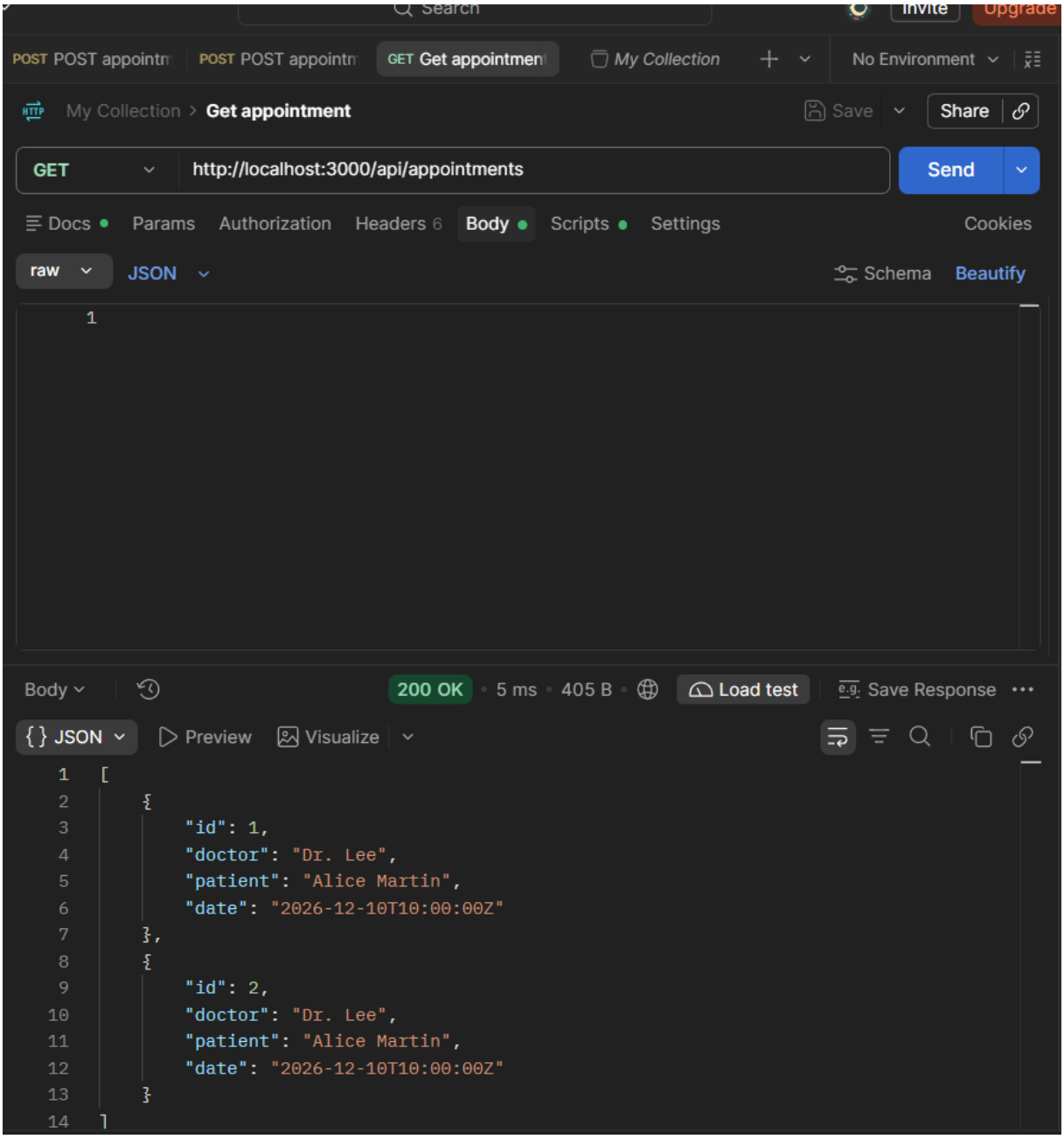
Appointments are stored in memory and reset when the server restarts. Date format validation and duplicate prevention are not implemented.

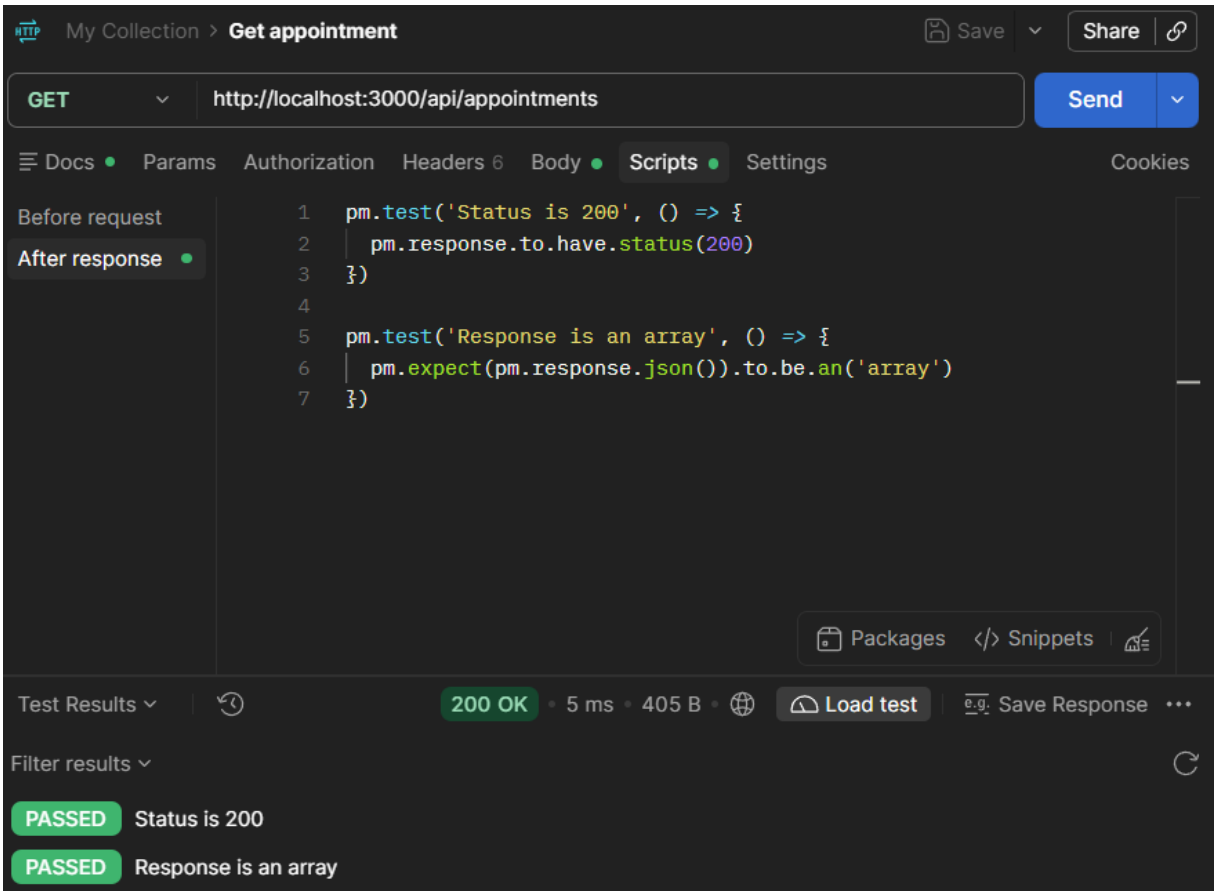
Testing and results

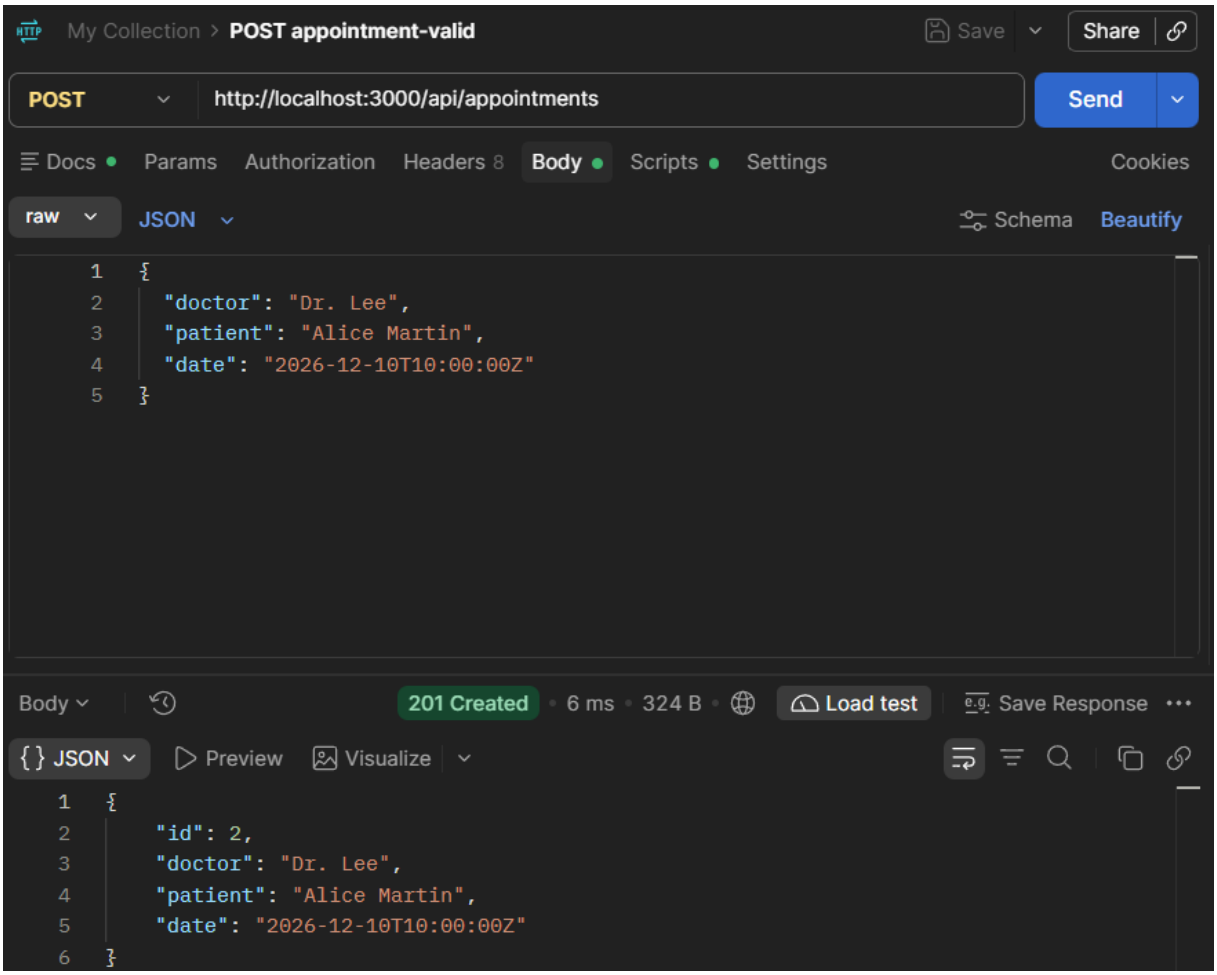
I added 20 automated integration tests for appointments. They check retrieval, creation, unique IDs, whitespace trimming and rejection of invalid requests. After integrating the patient feature from the shared branch, all 45 project tests passed and ESLint reported no errors.

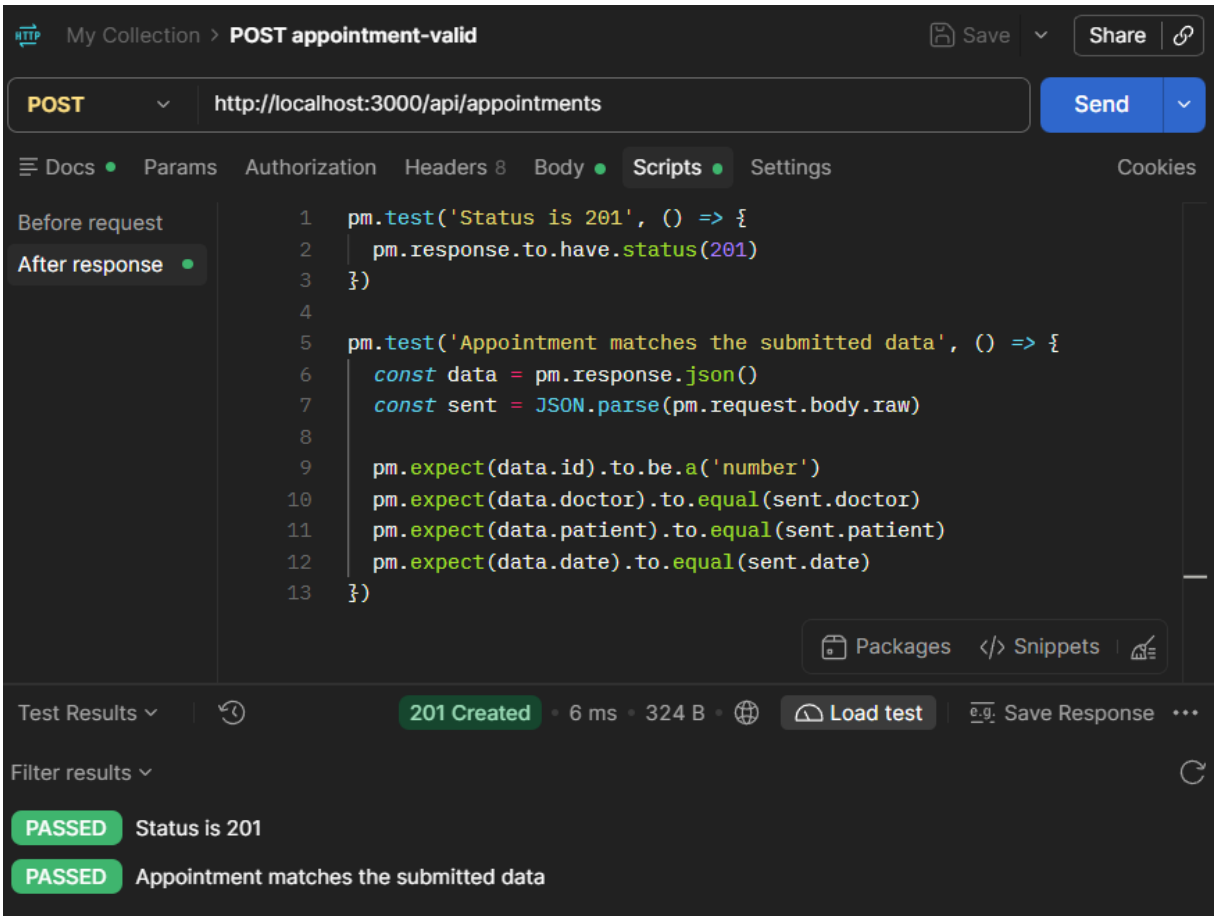
Global coverage reached **95.83% for lines, 95% for statements, 87.5% for functions and 78.12% for branches**, meeting the project thresholds.

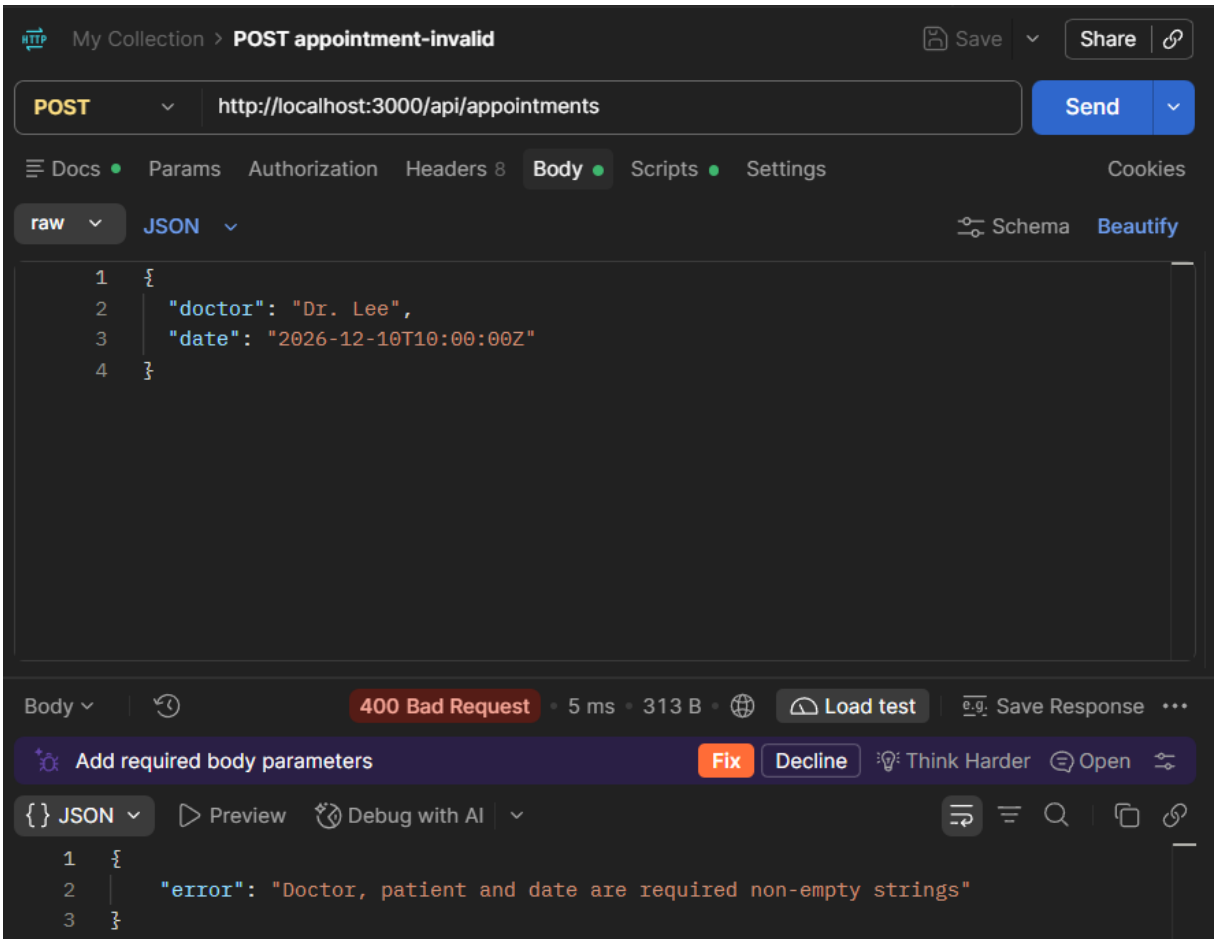
I also verified the API in Postman using one GET request, one valid POST and one invalid POST. All six Postman assertions passed. The collection was exported and committed under postman/.

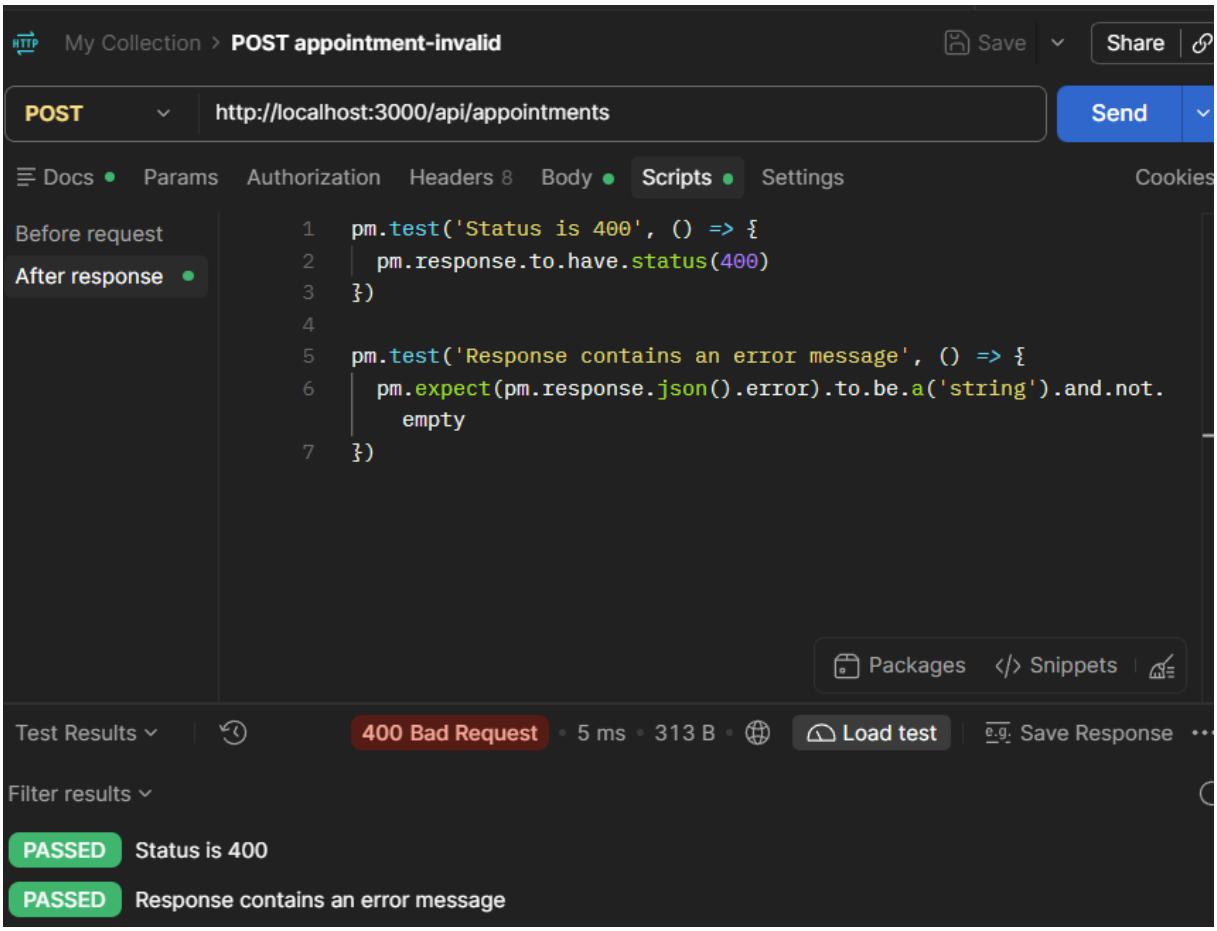

POST POST appointm
POST POST appointm
GET Get appointmen
My Collection
No Environment
My Collection > Get appointment
Save
Share
GET
http://localhost:3000/api/appointments
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
1
Body
200 OK
5 ms
405 B
Load test
Save Response
{ } JSON
Preview
Visualize
1 [
2 {
3 "id": 1,
4 "doctor": "Dr. Lee",
5 "patient": "Alice Martin",
6 "date": "2026-12-10T10:00:00Z"
7 },
8 {
9 "id": 2,
10 "doctor": "Dr. Lee",
11 "patient": "Alice Martin",
12 "date": "2026-12-10T10:00:00Z"
13 }
14]


My Collection > Get appointment
Save
Share
GET
http://localhost:3000/api/appointments
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Settings
Cookies
Before request
After response
1 pm.test('Status is 200', () => {
2 pm.response.to.have.status(200)
3 })
4
5 pm.test('Response is an array', () => {
6 pm.expect(pm.response.json()).to.be.an('array')
7 })
Packages
Snippets
Test Results
200 OK
5 ms
405 B
Load test
Save Response
Filter results
PASSED
Status is 200
PASSED
Response is an array


My Collection > POST appointment-valid
Save
Share
POST
http://localhost:3000/api/appointments
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
"doctor": "Dr. Lee",
"patient": "Alice Martin",
"date": "2026-12-10T10:00:00Z"
}
Body
201 Created
6 ms
324 B
Load test
Save Response
JSON
Preview
Visualize
{
"id": 2,
"doctor": "Dr. Lee",
"patient": "Alice Martin",
"date": "2026-12-10T10:00:00Z"
}

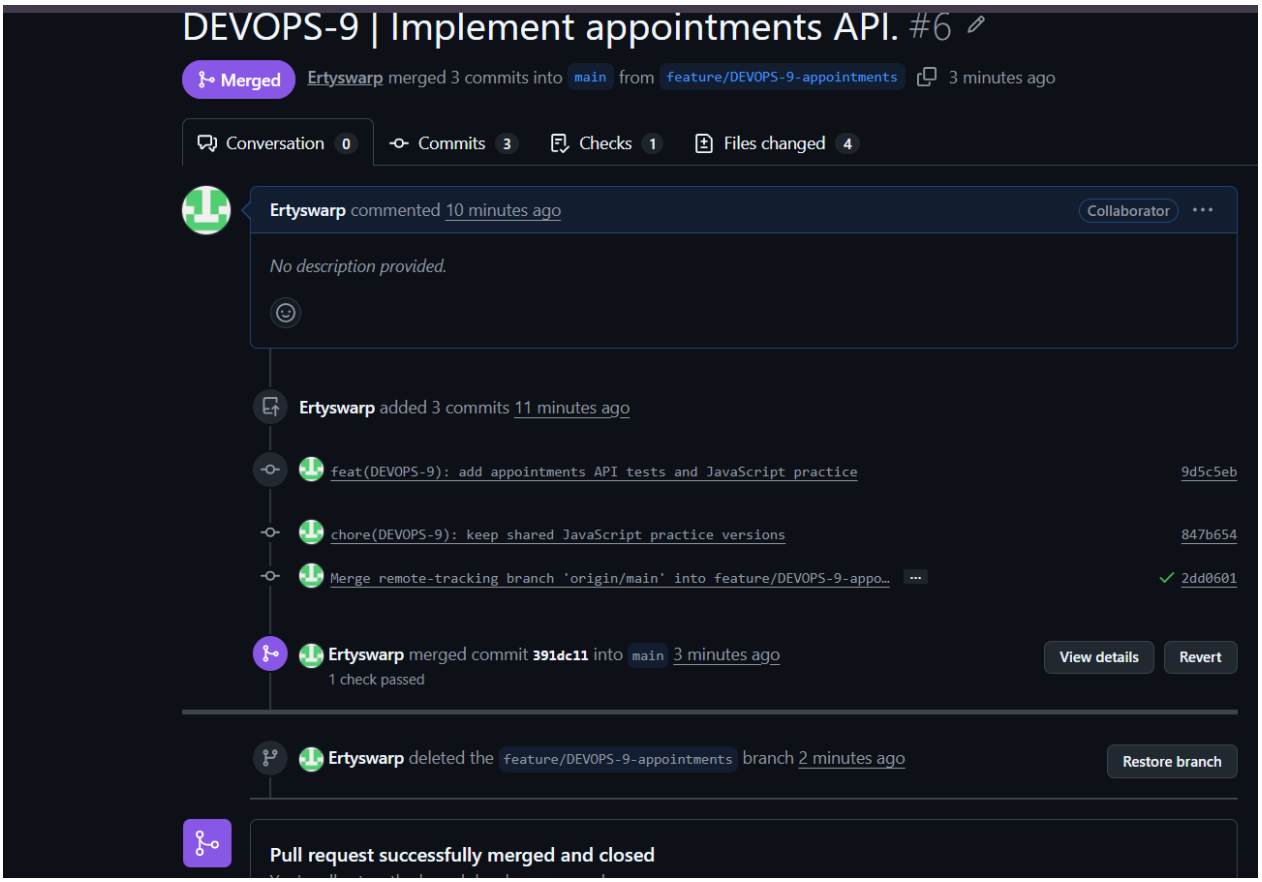

My Collection > POST appointment-valid
Save
Share
POST
http://localhost:3000/api/appointments
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
Before request
After response
pm.test('Status is 201', () => {
pm.response.to.have.status(201)
})
pm.test('Appointment matches the submitted data', () => {
const data = pm.response.json()
const sent = JSON.parse(pm.request.body.raw)
pm.expect(data.id).to.be.a('number')
pm.expect(data.doctor).to.equal(sent.doctor)
pm.expect(data.patient).to.equal(sent.patient)
pm.expect(data.date).to.equal(sent.date)
})
Packages
Snippets
Test Results
201 Created
6 ms
324 B
Load test
Save Response
Filter results
PASSED
Status is 201
PASSED
Appointment matches the submitted data


My Collection > POST appointment-invalid
Save
Share
POST
http://localhost:3000/api/appointments
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
"doctor": "Dr. Lee",
"date": "2026-12-10T10:00:00Z"
}
Body
400 Bad Request
5 ms
313 B
Load test
Save Response
Add required body parameters
Fix
Decline
Think Harder
Open
JSON
Preview
Debug with AI
{
"error": "Doctor, patient and date are required non-empty strings"
}


My Collection > POST appointment-invalid
Save
Share
POST
http://localhost:3000/api/appointments
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
Before request
After response
pm.test('Status is 400', () => {
pm.response.to.have.status(400)
})
pm.test('Response contains an error message', () => {
pm.expect(pm.response.json().error).to.be.a('string').and.not.
empty
})
Packages
Snippets
Test Results
400 Bad Request
5 ms
313 B
Load test
Save Response
Filter results
PASSED
Status is 400
PASSED
Response contains an error message

Git and Jira workflow

I tracked my work in Jira Story **DEVOPS-9** and developed it on feature/DEVOPS-9-appointments. I opened Pull Request **#6** and resolved conflicts involving the shared JavaScript practice files by retaining the versions from main. After successful CI validation, the PR was merged and the Story was marked Done.



DEVOPS-5 / DEVOPS-9

Implement appointments API

Description

As a hospital staff member,
I want to create and view appointments,
so that I can schedule visits between doctors and patients.

Acceptance criteria:

- GET /api/appointments returns an array with HTTP 200.
- POST /api/appointments accepts doctor, patient and date.
- A valid appointment receives a unique ID and HTTP 201.
- Missing or empty required fields return HTTP 400.
- A rejected request does not create an appointment.
- Automated tests pass and coverage meets the project thresholds.
- The endpoints are verified with Postman.

For this lab, appointments are stored in memory and reset when the server restarts.

Terminé | Done

Details

Assignee	Zixuan Qian
Parent	DEVOPS-5 API Core Devel...
Sprint	Add sprint
Labels	Add labels
Due date	Add due date
Team	Add team
Start date	Add date
Story point estimate	Add story points
Fix versions	None

```
>> npm.cmd test -- --coverage
 ❯ test/appointments.test.js [queued]
 ✓ test/unit/appInfo.test.js (3 tests) 11ms
 ✓ test/unit/errorHandler.test.js (2 tests) 6ms
 ✓ test/unit/status.test.js (4 tests) 8ms


 ✓ test/boom.test.js (1 test) 91ms
 ✓ test/health.test.js (1 test) 90ms
 ✓ test/version.test.js (2 tests) 116ms
 ✓ test/info.test.js (2 tests) 120ms
 ✓ test/patients.test.js (10 tests) 276ms
 ✓ test/appointments.test.js (20 tests) 662ms

 Test Files  9 passed (9)
      Tests  45 passed (45)
   Start at  17:07:47
   Duration  1.55s (transform 892ms, setup 0ms, import 3.14s, tests 1.38s, environment 1ms)

 % Coverage report from v8
-----------------------|---------|----------|---------|---------|-------------------
File                   | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-----------------------|---------|----------|---------|---------|-------------------
All files              |      95 |    78.12 |    87.5 |   95.83 |
 src                   |    82.6 |     37.5 |      50 |   84.21 |
  app.js               |      95 |       50 |   66.66 |     100 | 30-36
  index.js             |       0 |        0 |       0 |       0 | 8-10
 src/routes/auto       |     100 |    85.71 |     100 |     100 |
  appointments.route.js |     100 |    83.33 |     100 |     100 | 16
  boom.route.js        |     100 |      100 |     100 |     100 |
  info.route.js        |     100 |      100 |     100 |     100 |
  patients.route.js    |     100 |     87.5 |     100 |     100 | 18
  version.route.js     |     100 |      100 |     100 |     100 |
 src/utils             |     100 |      100 |     100 |     100 |
  appInfo.js           |     100 |      100 |     100 |     100 |
  errorHandler.js      |     100 |      100 |     100 |     100 |
  status.js            |     100 |      100 |     100 |     100 |
-----------------------|---------|----------|---------|---------|-------------------
```

The results from the practice exercises

```
PS C:\SoftwareEngineering\lab5> node js-practice.js
>> node main.js
>> npm.cmd run lint
Welcome to Central Hospital! We have 12 doctors currently available.
Central Hospital is currently closed. Please come back later.
Doctor #1: Dr. Lee
Doctor #2: Dr. Khan
Doctor #3: Dr. Martin
Doctor #4: Dr. Sarah
Doctor #5: Dr. Amir
Doctors count: 6
Doctor found
Doctor not found
Dr. Lee
2
Alice Martin is treated by Dr. Lee (Cardiology).
{
  name: 'Alice Martin',
  age: 34,
  conditions: [ 'diabetes', 'hypertension', 'anxiety' ],
  doctor: { name: 'Dr. Lee', speciality: 'Cardiology' }
}
Older than 30: [ { id: 1, name: 'Alice', age: 34 }, { id: 2, name: 'John', age: 45 } ]
Updated patients: [
  { id: 1, name: 'Alice', age: 34 },
  { id: 2, name: 'John', age: 45 },
  { id: 3, name: 'Marie', age: 29 },
  { id: 4, name: 'Zoe', age: 51 }
]
{ doctor: 'Dr. Lee', patient: 'Alice', date: '2025-03-15' }
Error: Missing required fields
Fetching hospital data...
Data loaded: { name: 'Central Hospital', uptime: 1234 }
Updated doctors: [ 'Dr. Lee', 'Dr. Khan', 'Dr. Martin' ]

> devops-lab-student-project@1.0.0 lint
> eslint .
```